

I am preparing for my semester exam please provide answers in human written the answers is not long and not short, Answers only medium, subject related to Cyber law & Security and Answers in easy word. For 5-6 marks .

Sure. Send the questions/topics from your Cyber Law & Security subject, and I will provide:

- Medium-length answers (perfect for 5–6 marks)
- Human-written style
- Easy language
- Exam-ready format
- Important points included
- No unnecessary long explanation

You can send:

- One question
- Multiple questions
- Or even photos of questions paper/notes.



Q.1 what are the major types of threats and attacks on information system? Explain How organization can assess damages caused by such attacks. give me answers upto 6 points.

Major Types of Threats and Attacks on Information System

Information systems face many types of threats and attacks that can damage data, software, hardware, and network security. Some major threats are explained below:

1. Malware Attack

Malware means harmful software such as viruses, worms, ransomware, and spyware. It enters the system and damages files, steals data, or slows down system performance.

2. Phishing Attack

In phishing, attackers send fake emails or messages to trick users into sharing passwords, bank details, or confidential information.

3. Denial of Service (DoS) Attack

In this attack, hackers send a huge amount of traffic to a server or website so that genuine users cannot access the service.

4. Password Attack

Attackers try to steal or guess passwords using methods like brute force attack, keylogging, or password cracking tools.

5. Insider Threat

Sometimes employees or authorized users misuse their access rights and leak important company information intentionally or accidentally.

6. Network Attack

Hackers attack computer networks using methods such as hacking, packet sniffing, or man-in-the-middle attacks to steal or modify data.

How Organizations Can Assess Damages Caused by Attacks

Organizations can measure and assess the damage caused by cyber attacks in the following ways:

1. Financial Loss Assessment

The organization calculates losses related to stolen money, recovery cost, legal penalties, and business interruption.

2. Data Loss Evaluation

Companies check how much important or confidential data has been deleted, leaked, or corrupted during the attack.

3. System Performance Analysis

Technical teams examine whether systems, servers, or networks are damaged or slowed down after the attack.

4. Reputation Damage Check

Organizations analyze customer trust, public image, and market reputation affected due to the cyber attack.

5. Security Audit and Investigation

Experts perform security audits and forensic analysis to identify the source, impact, and severity of the attack.

6. Recovery Time Measurement

Organizations estimate how much time and resources are required to restore normal operations and recover affected systems.



Define Confidentiality, Integrity and availability in information system. Why are these considered core principles of the CIA triad? for (5-6) marks

Confidentiality, Integrity and Availability in Information System

The CIA Triad is a basic security model used to protect information systems and data. It consists of three important principles: Confidentiality, Integrity, and Availability.

1. Confidentiality

Confidentiality means protecting information from unauthorized access or disclosure. Only authorized users are allowed to view sensitive data. Passwords, encryption, and authentication methods are used to maintain confidentiality.

Example: Bank account details should only be accessible to the account holder and authorized bank staff.

2. Integrity

Integrity means maintaining the accuracy and correctness of data. Information should not be changed, deleted, or modified by unauthorized persons. Security controls help ensure that data remains trustworthy.

Example: Student marks stored in a database should not be altered illegally.

3. Availability

Availability means that information and system resources should be accessible to authorized users whenever needed. Systems should work properly without unnecessary interruption.

Example: An online shopping website should remain available for customers 24×7.

Why These are Core Principles of CIA Triad

The CIA triad is considered the foundation of information security because it helps organizations protect their systems and data effectively.

1. Protects Sensitive Information

Confidentiality prevents unauthorized users from accessing private data.

2. Ensures Data Accuracy

Integrity helps maintain correct and reliable information.

3. Provides Continuous Access

Availability ensures that users can access systems and services without downtime.

4. Builds Trust and Security

These principles increase customer trust and improve system reliability.

5. Helps in Cyber Security Management

Organizations use the CIA triad to design security policies, risk management, and protection mechanisms.



Q.3 Explain different types of information system with suitable examples. for 5 marks

Different Types of Information System with Examples

An Information System is a system that collects, processes, stores, and distributes information to support decision-making and management in an organization. Different types of information systems are explained below:

1. Transaction Processing System (TPS)

TPS is used to handle daily routine transactions of an organization quickly and accurately.

Example: ATM transactions, railway reservation systems, and billing systems in supermarkets.

2. Management Information System (MIS)

MIS provides summarized reports and information to managers for planning and decision-making.

Example: Monthly sales reports and employee performance reports.

3. Decision Support System (DSS)

DSS helps managers in solving complex problems and making better decisions by analyzing data.

Example: Weather forecasting systems and business analysis software.

4. Executive Information System (EIS)

EIS provides important business information to top-level executives in a simple and graphical form.

Example: Dashboard showing company profit, growth, and market performance.

5. Office Automation System (OAS)

OAS helps employees perform office-related work efficiently using computer software.

Example: MS Word, email systems, video conferencing, and spreadsheet software.

6. Expert System

Expert systems use artificial intelligence to provide expert advice and solutions in specific fields.

Example: Medical diagnosis systems and online customer support chatbots.

These information systems improve communication, decision-making, efficiency, and overall business performance.



Q.4 Discuss the changing nature of information system in the era of globalization and digital transformation. for (5-6) marks

Changing Nature of Information System in the Era of Globalization and Digital Transformation

The nature of Information System has changed rapidly due to globalization and digital transformation. Modern organizations now use advanced technologies to improve communication, business operations, and decision-making at a global level.

1. Global Connectivity

Information systems now connect people and organizations across different countries through the internet and cloud technology. Businesses can communicate and operate globally in real time.

2. Shift from Manual to Digital Systems

Traditional paper-based work has been replaced by digital systems. Organizations use online platforms, databases, and software applications for faster and more accurate work.

3. Use of Cloud Computing

Cloud computing allows organizations to store and access data from anywhere. It reduces hardware cost and improves flexibility and scalability.

4. Growth of E-Commerce and Online Services

Businesses now provide products and services online through websites and mobile applications. Customers can shop, pay bills, and access services digitally.

5. Increased Importance of Cyber Security

As digital data and online transactions increase, protecting information systems from cyber attacks has become very important. Organizations use security tools, encryption, and firewalls for protection.

6. Use of Advanced Technologies

Modern information systems use technologies like Artificial Intelligence (AI), Big Data, and the Internet of Things (IoT) to improve automation and decision-making.

Thus, globalization and digital transformation have made information systems faster, smarter, more connected, and more important for business success.



Q.5 Describe the importance of authentication service and organizational security applications. for 5 marks

Importance of Authentication Service and Organizational Security Applications

Authentication service and security applications are important parts of Cyber Security. They help organizations protect data, systems, and networks from unauthorized access and cyber attacks.

1. User Verification

Authentication services verify the identity of users before allowing access to systems or data. Common methods include passwords, OTPs, biometric scans, and smart cards.

2. Protection of Sensitive Information

Authentication prevents unauthorized persons from accessing confidential information such as employee records, financial data, and customer details.

3. Prevention of Cyber Attacks

Security applications like firewalls, antivirus software, and intrusion detection systems protect organizations from malware, hacking, and phishing attacks.

4. Secure Access Control

Organizations use authentication systems to provide different access levels to employees based on their roles and responsibilities.

5. Maintaining Data Integrity

Security applications help protect data from unauthorized modification or deletion, ensuring information remains accurate and reliable.

6. Building Trust and Business Continuity

Strong authentication and security systems increase customer trust and help organizations continue operations safely without major disruptions.

Therefore, authentication services and organizational security applications are essential for maintaining confidentiality, integrity, and availability of information systems.



Q.6 what is information assurance ? Explain cyber security and security risk analysis in detail. for 5 marks

Information Assurance, Cyber Security and Security Risk Analysis

Information Assurance

Information Assurance refers to the process of protecting information systems and data from unauthorized access, modification, disclosure, or destruction. It ensures that information remains secure, accurate, and available whenever needed.

The main goal of information assurance is to maintain:

- Confidentiality
- Integrity
- Availability
- Authentication
- Non-repudiation

It uses security policies, technologies, and risk management techniques to protect information.

Cyber Security

Cyber Security is the practice of protecting computers, networks, software, and data from cyber attacks and unauthorized access.

Importance of Cyber Security

1. Protects sensitive and personal data.
2. Prevents hacking, malware, and phishing attacks.
3. Ensures safe online transactions and communication.
4. Protects organizational reputation and customer trust.
5. Helps maintain smooth business operations.

Common Cyber Security Tools

- Antivirus software
- Firewalls
- Encryption
- Intrusion Detection Systems (IDS)

- Multi-factor authentication
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Security Risk Analysis

Security risk analysis is the process of identifying, evaluating, and managing security risks that may harm an organization's information system.

Steps in Security Risk Analysis

1. **Identify Threats** – Detect possible cyber threats and vulnerabilities.
2. **Assess Risks** – Analyze the probability and impact of attacks.
3. **Evaluate Existing Security** – Check whether current security controls are effective.
4. **Apply Risk Control Measures** – Implement firewalls, encryption, backup systems, and security policies.
5. **Monitor and Review** – Regularly review risks and update security measures.

Security risk analysis helps organizations reduce losses, improve protection, and maintain information security effectively.



Q.7 Discuss the security challenges in mobile and wireless computing, explain backup Archival storage and secure disposal of data in information security. for 5 marks

Security Challenges in Mobile and Wireless Computing

Mobile Computing and wireless technologies provide flexibility and easy communication, but they also create many security challenges.

1. Unauthorized Access

Hackers may access mobile devices or wireless networks without permission if security settings are weak.

2. Data Theft and Device Loss

Mobile phones and laptops can be lost or stolen, which may expose confidential data.

3. Malware and Virus Attacks

Mobile devices are vulnerable to malicious applications, spyware, and viruses downloaded from unsafe sources.

4. Unsecured Wi-Fi Networks

Public Wi-Fi networks are often insecure, allowing attackers to intercept user data and passwords.

5. Phishing and Fake Applications

Cyber criminals use fake messages and applications to steal personal and financial information.

6. Weak Authentication

Simple passwords and lack of multi-factor authentication increase security risks in mobile computing.

Backup, Archival Storage and Secure Disposal of Data

1. Backup

Backup means creating copies of important data so that it can be restored if data is lost, corrupted, or attacked by malware.

Importance of Backup:

- Prevents permanent data loss
- Helps recover data after system failure
- Supports business continuity

Example: Cloud backup and external hard drives.

2. Archival Storage

Archival storage refers to storing old or less frequently used data for a long period in a secure manner.

Importance of Archival Storage:

- Preserves important records and documents
- Helps meet legal and business requirements
- Saves storage space in active systems

Example: Government records and old company files stored in cloud archives.

3. Secure Disposal of Data

Secure disposal means permanently deleting or destroying data so that unauthorized users cannot recover it.

Methods of Secure Disposal:

- Data wiping
- Disk formatting
- Physical destruction of storage devices

Importance:

- Prevents data leakage
- Protects confidential information
- Reduces risk of cyber crimes

Thus, backup, archival storage, and secure disposal are important practices in Information Security for protecting organizational data.



Q.8 Explain the role of Internet and web services in the development of the information system. for 5 marks

Role of Internet and Web Services in the Development of Information System

The Internet and web services play an important role in the growth and development of modern Information System. They help organizations communicate, share data, and provide services quickly and efficiently.

1. Global Communication

The internet allows fast communication between people and organizations across the world through email, video conferencing, and messaging services.

2. Easy Access to Information

Web services help users access information anytime and from anywhere using websites and cloud platforms.

3. Support for E-Commerce

Internet technology enables online shopping, online banking, ticket booking, and digital payment systems, making business operations easier.

4. Data Sharing and Integration

Web services allow different applications and systems to exchange data and work together smoothly through APIs and cloud services.

5. Remote Working and Collaboration

Employees can work remotely and collaborate using online tools such as Google Workspace, Zoom, and Microsoft Teams.

6. Improved Decision Making

Organizations use internet-based information systems to collect real-time data, analyze information, and make better business decisions.

7. Cost and Time Efficiency

Internet and web services reduce paperwork, save time, lower communication costs, and improve overall productivity.

Thus, the internet and web services have made information systems faster, more connected, and more efficient in modern organizations.



Q. 9 Explain the need for distributed security information system with suitable examples. for 5 marks

Need for Distributed Security Information System with Suitable Examples

A Distributed Information System is a system in which data, applications, and resources are distributed across multiple computers or locations connected through a network. Distributed security is needed to protect these systems from cyber threats and unauthorized access.

Need for Distributed Security Information System

1. Protection of Shared Data

In distributed systems, data is shared between many computers and users. Security is needed to prevent unauthorized access and data leakage.

2. Secure Communication

Data travels through networks in distributed systems. Security mechanisms like encryption help protect information during transmission.

3. User Authentication

Distributed systems require proper authentication to ensure that only authorized users can access resources and services.

4. Prevention of Cyber Attacks

Security systems protect distributed networks from hacking, malware, phishing, and denial-of-service attacks.

5. Data Availability and Reliability

Distributed security ensures continuous access to data and services even if one server or system fails.

6. Protection of Remote Access

Many organizations allow employees to work remotely. Security tools like VPNs and firewalls protect remote connections.

Suitable Examples

1. Online Banking System

Bank branches in different cities use distributed systems to access customer accounts securely.

2. Cloud Computing Services

Cloud platforms like Google Drive and Dropbox store data on multiple servers and use strong security measures.

3. E-Commerce Websites

Websites such as Amazon and Flipkart use distributed systems to manage users, payments, and product databases securely.

Thus, distributed security information systems are important for maintaining confidentiality, integrity, and availability of information in modern network environments.



Q.10 Compare information security and privacy . How do security metrics help in evaluating the effectiveness of security Program? with heading features in comparing for 5 marks

Comparison between Information Security and Privacy

Features	Information Security	Privacy
Meaning	Information security protects data and systems from unauthorized access, attacks, and damage.	Privacy protects personal and confidential information from unnecessary collection or misuse.
Main Focus	Focuses on protection of data, networks, and systems.	Focuses on the rights of individuals over their personal information.
Objective	To maintain confidentiality, integrity, and availability of data.	To control how personal information is collected, shared, and used.
Methods Used	Firewalls, encryption, antivirus, authentication, and backup systems.	Privacy policies, consent, data protection laws, and access control.
Concern	Prevents cyber attacks and data breaches.	Prevents misuse of personal information.
Example	Protecting company servers from hackers.	Protecting users' personal details on social media platforms.

Role of Security Metrics in Evaluating Security Programs

Information Security metrics are measurements used to evaluate how effective a security program is in protecting an organization.

1. Measures Security Performance

Security metrics help organizations check whether security controls and policies are working properly.

2. Identifies Weaknesses

Metrics help detect vulnerabilities, weak systems, and areas that need improvement.

3. Supports Decision Making

Management uses security reports and metrics to make better security-related decisions.

4. Reduces Security Risks

Regular monitoring through metrics helps reduce cyber threats and prevent future attacks.

5. Ensures Compliance

Security metrics help organizations follow security standards, laws, and regulations.

6. Improves Overall Security

Continuous evaluation helps organizations strengthen their security programs and improve protection of information systems.

